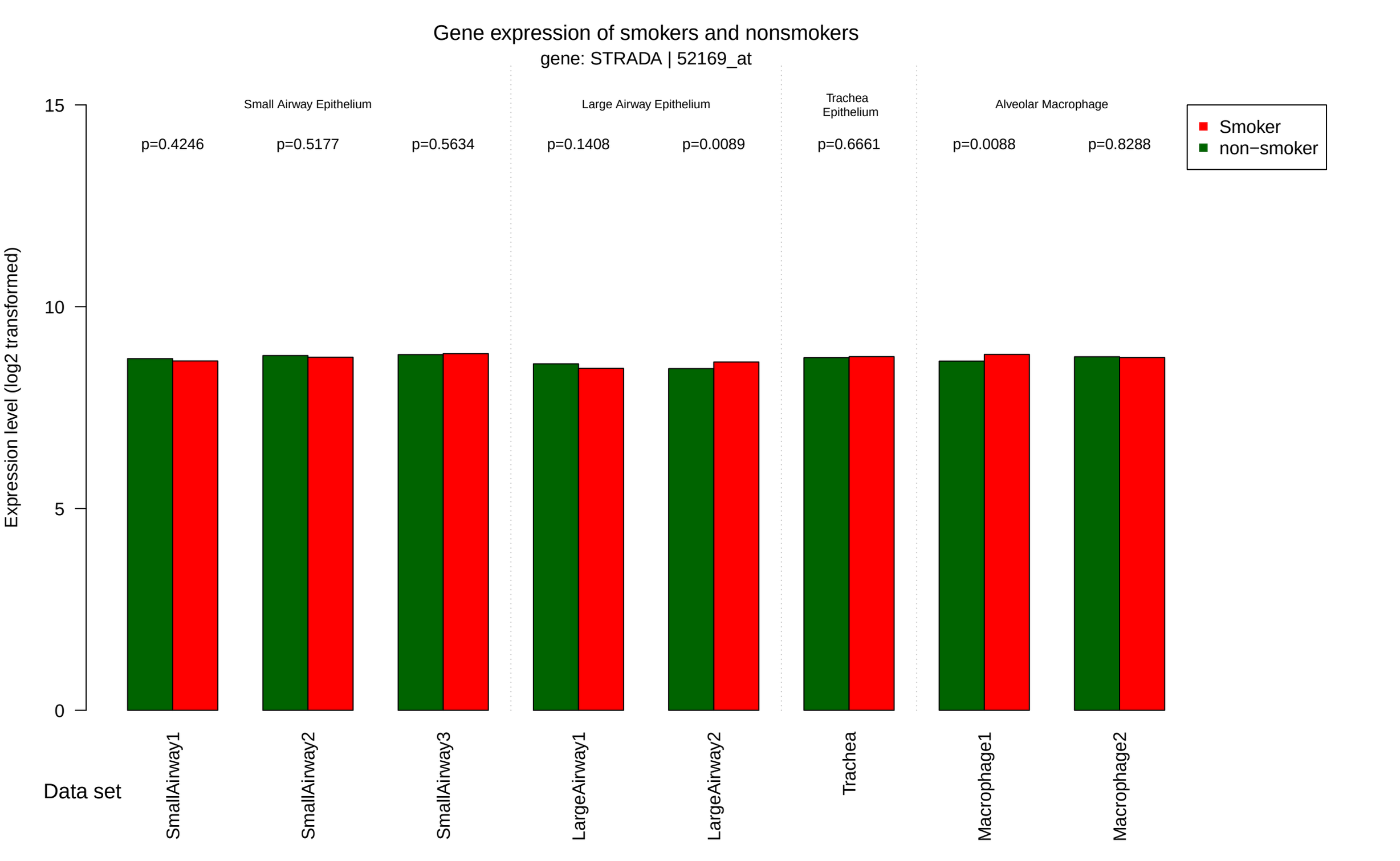
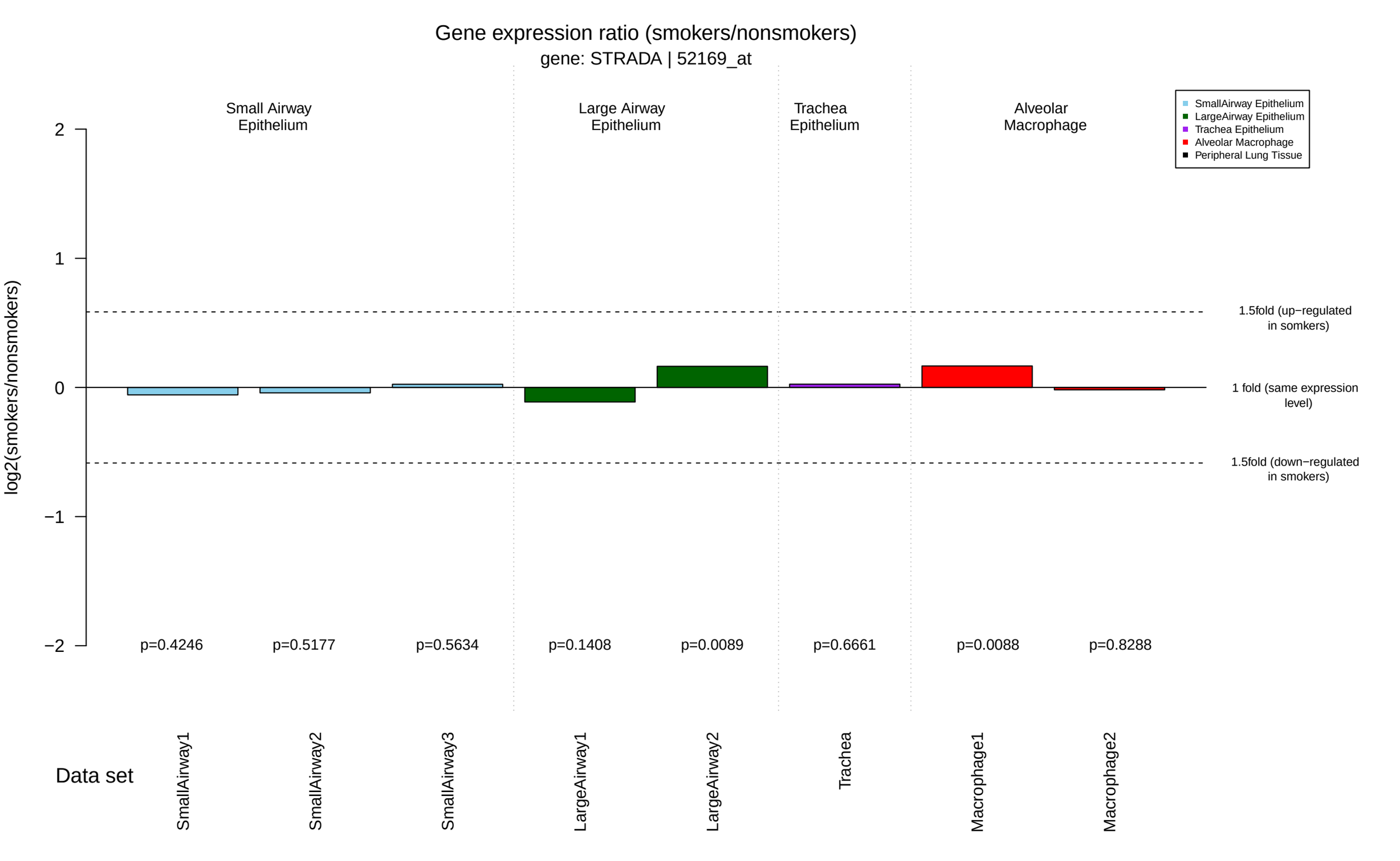
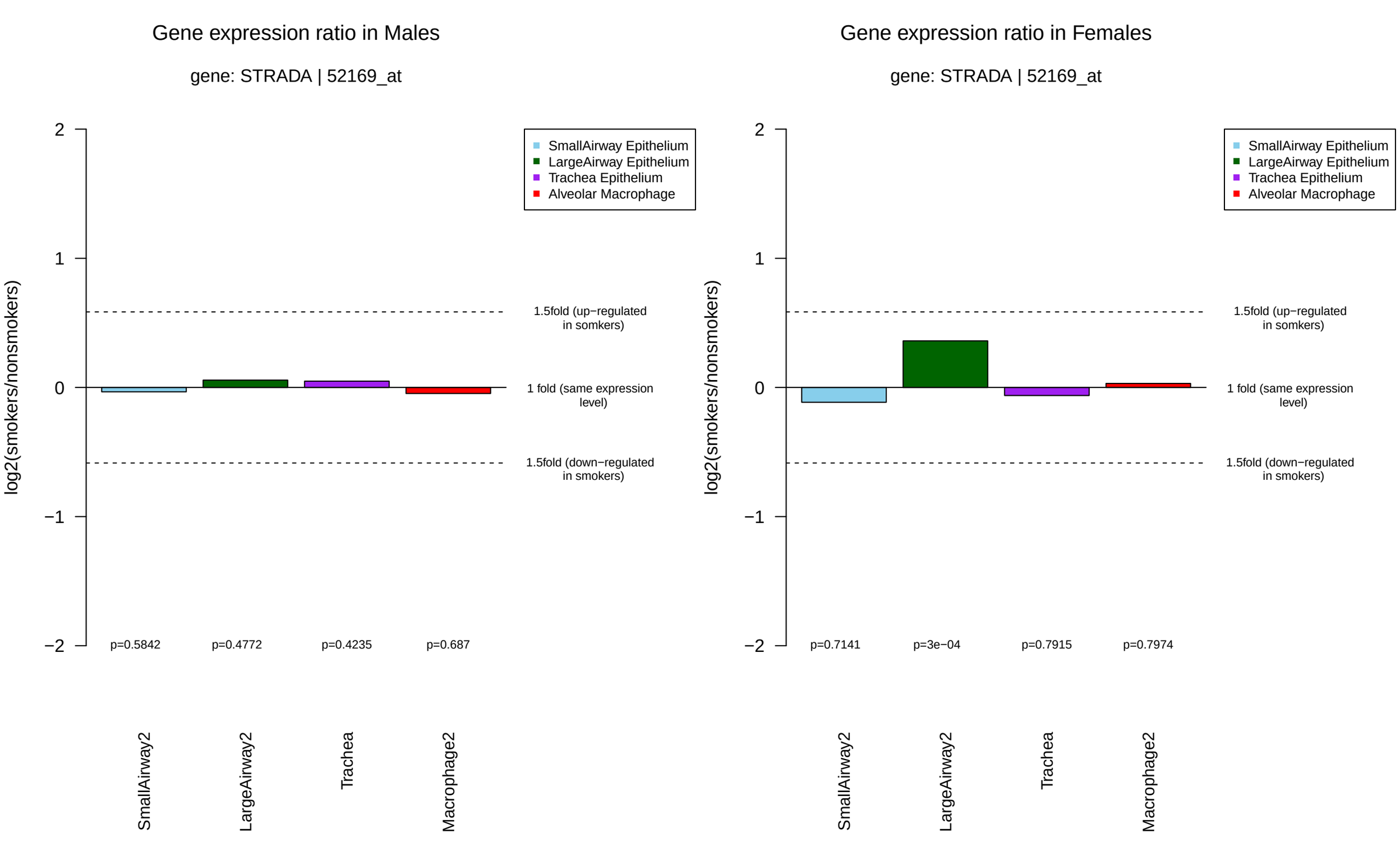
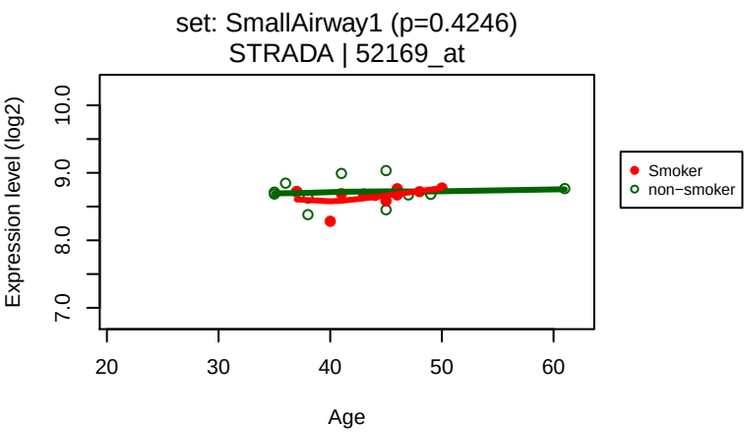
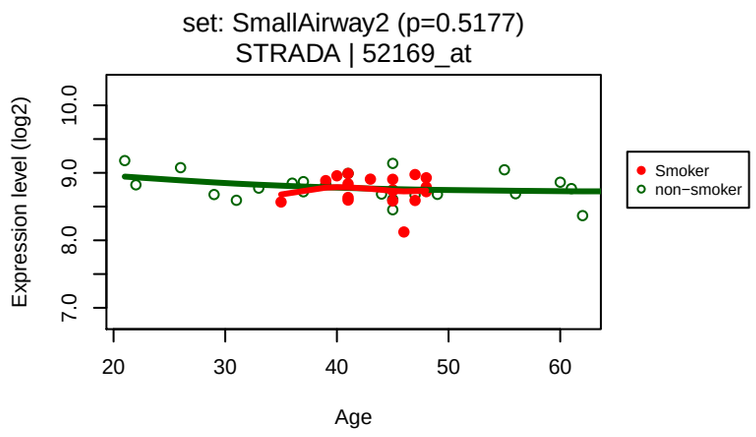
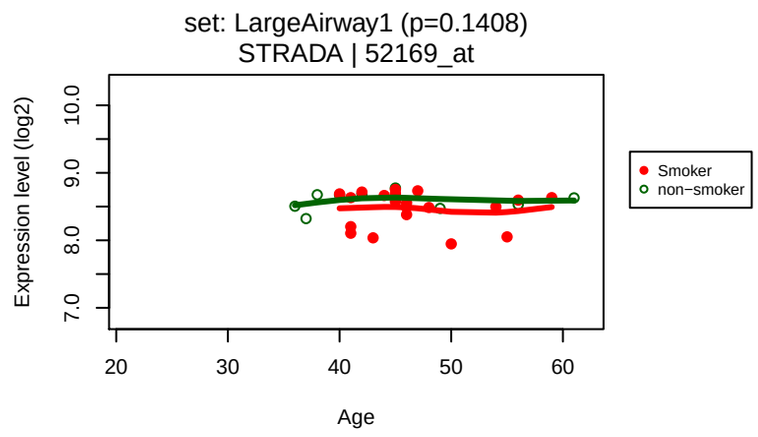

Gene expression of smokers and nonsmokers
gene: STRADA | 52169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.4246
p=0.5177
p=0.5634
p=0.1408
p=0.0089
p=0.6661
p=0.0088
p=0.8288
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

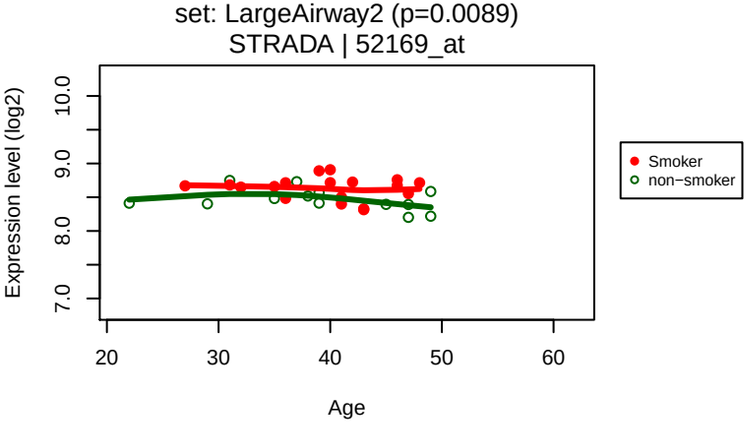

Gene expression ratio (smokers/nonsmokers)
gene: STRADA | 52169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4246
p=0.5177
p=0.5634
p=0.1408
p=0.0089
p=0.6661
p=0.0088
p=0.8288
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

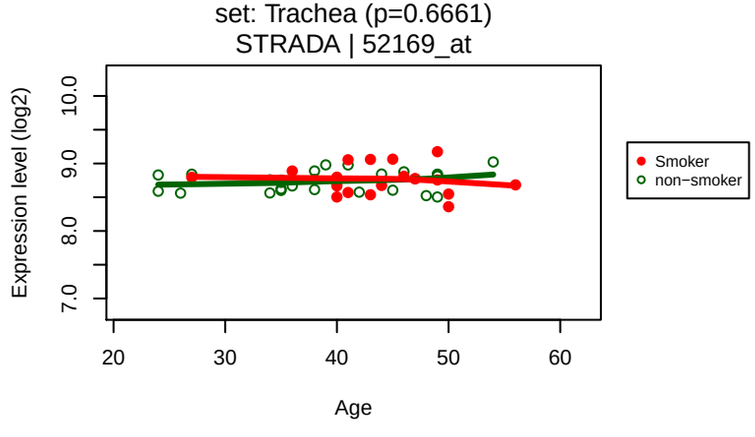

Gene expression ratio in Males
gene: STRADA | 52169_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5842
p=0.4772
p=0.4235
p=0.687
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: STRADA | 52169_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7141
p=3e−04
p=0.7915
p=0.7974
SmallAirway2
LargeAirway2
Trachea
Macrophage2

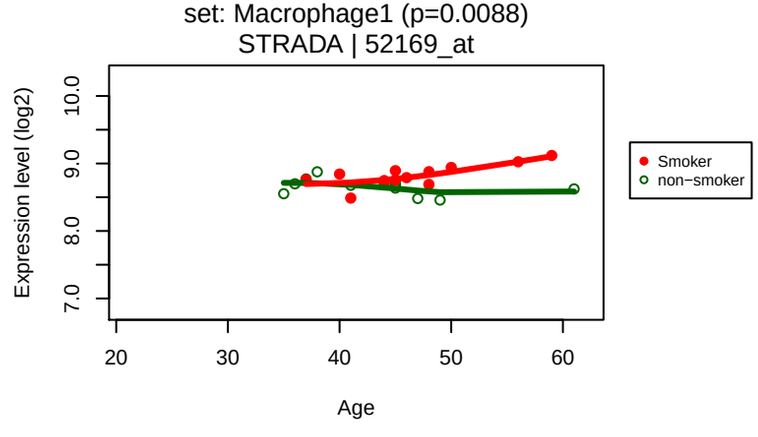

set: SmallAirway1 (p=0.4246)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

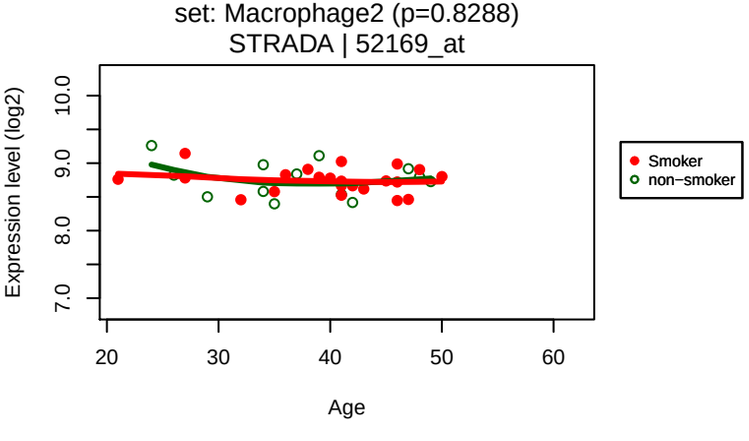

set: SmallAirway2 (p=0.5177)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker

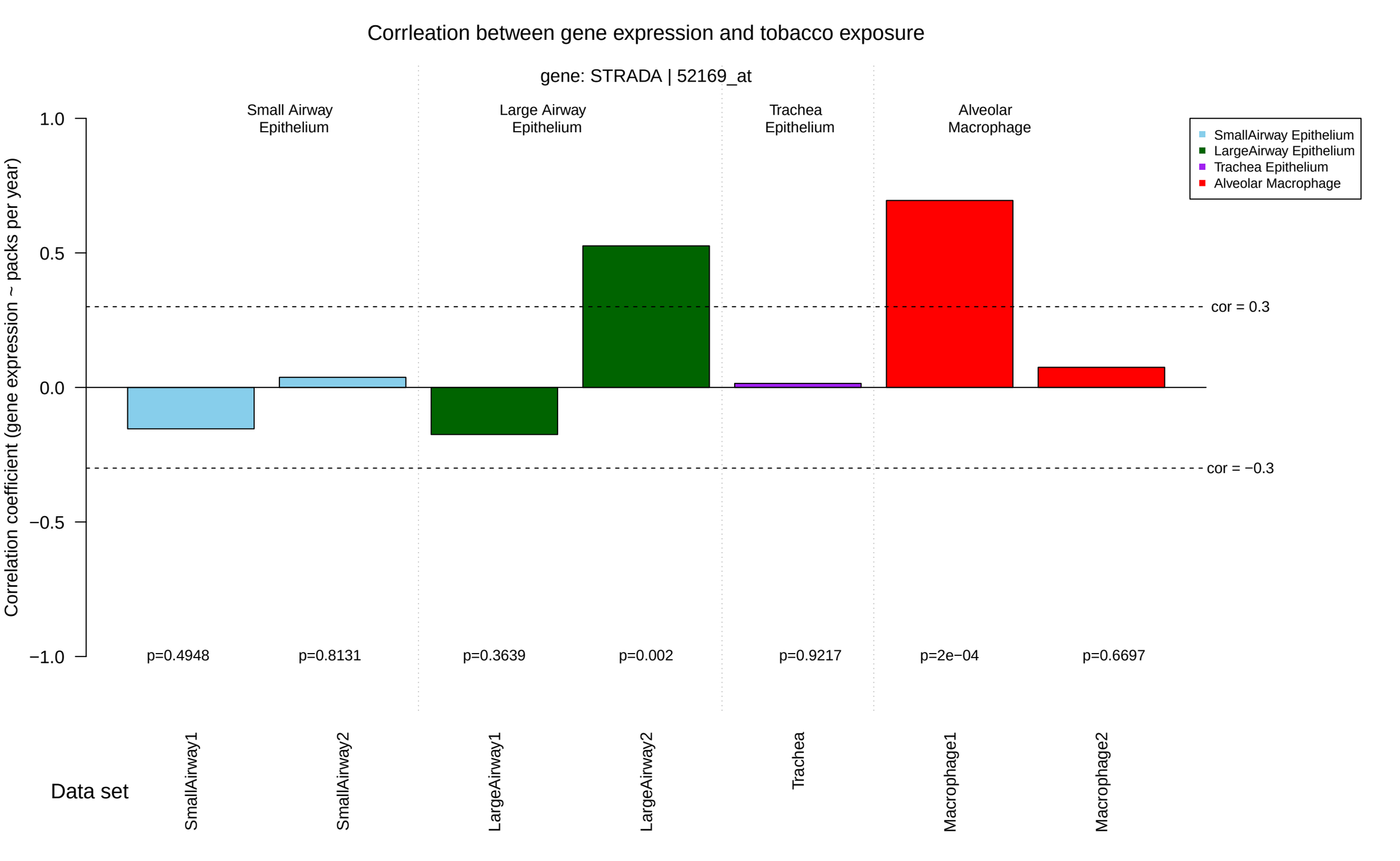

set: LargeAirway1 (p=0.1408)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0089)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.6661)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.0088)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8288)
STRADA | 52169_at
Expression level (log2)
7.0
8.0
9.0
10.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: STRADA | 52169_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.0
0.5
0.0
−0.5
−1.0
Correlation coefficient (gene expression ~ packs per year)
cor = 0.3
cor = −0.3
p=0.4948
p=0.8131
p=0.3639
p=0.002
p=0.9217
p=2e−04
p=0.6697
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2